Lesson 10.5 - Hypothesis Testing Problems - Z Test and T Statistics

> **Video:** Hypothesis Testing Problems - Z Test & T Statistics - One & Two Tailed Tests by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

Procedure (critical-value method)

1. State H_0 (status quo, with the equality) and H_a (the claim being tested).
2. Decide on the tails from H_a: $\neq \rightarrow$ two-tailed, $< \rightarrow$ left, $> \rightarrow$ right.
3. Find the **critical value** from α (z-table, or t-table with df $= n - 1$).
4. Compute the **test statistic**: $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, treated as z if $n \geq 30$ or as t if $n < 30$ and σ is unknown.
5. If the statistic is in the rejection region, **reject** H_0. Otherwise, **fail to reject** H_0.

Example 1 - two-tailed z-test (a bottling machine)

A machine is supposed to dispense **80 mL**. An employee thinks the mean is not 80 mL. A sample of **40** bottles has $\bar{x} = 78$ mL and $s = 2.5$ mL.

(a) Hypotheses

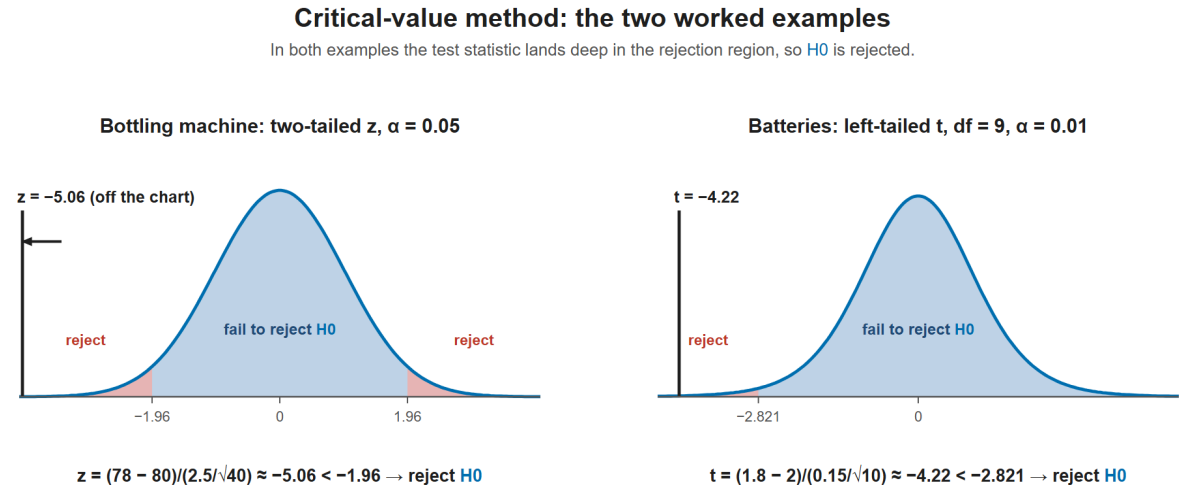
$$H_0 : \mu = 80 \qquad H_a : \mu \neq 80$$

(b) Is there evidence at 95% confidence that the machine is off?

- H_a uses $\neq$, so this is **two-tailed**: 0.025 in each tail, critical values $z = \pm 1.96$.
- $n = 40 \geq 30$, so use z (with s in place of σ).

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{78 - 80}{2.5/\sqrt{40}} = \frac{-2}{0.39528} \approx -5.06$$

$-5.06 < -1.96$, so the statistic is far into the left rejection region. **Reject H_0.** At the 5% significance level there is strong evidence that the machine does not dispense 80 mL on average.



Both worked examples. The bottling statistic (−5.06) is so extreme it lies off the chart. The battery test uses a t-distribution with 9 degrees of freedom, so its left critical value is −2.821, not a z value.

Example 2 - left-tailed t-test (car batteries)

The company says its batteries last **2 or more years** on average. An engineer thinks it's less. A sample of **10** batteries has $\bar{x} = 1.8$ years and $s = 0.15$ years.

(a) Hypotheses

$$H_0 : \mu \geq 2 \qquad H_a : \mu < 2$$

(b) Test at 99% confidence

- H_a uses $<$, so this is **left-tailed**, with $\alpha = 0.01$ entirely in the left tail.
- $n = 10 < 30$ and σ is unknown, so use the **t-distribution** with df $= 10 - 1 = 9$.
- One-tail t-table, df = 9, $\alpha = 0.01$: $t = 2.821$, so the critical value is -2.821 (left side).

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1.8 - 2}{0.15/\sqrt{10}} = \frac{-0.2}{0.047434} \approx -4.22$$

$-4.22 < -2.821$, which is in the rejection region. **Reject H_0.** At the 1% significance level there is enough evidence that the mean battery life is less than 2 years.

> **Wording:** the video sometimes says "accept the null hypothesis". The standard phrasing is **"fail to reject H_0"**. Not rejecting doesn't prove H_0 is true.

Key takeaways

- H_a decides the tails. n and whether σ is known decide z or t.
- Test statistic = (sample mean − hypothesized mean) ÷ (SD / √n).
- If the statistic is beyond the critical value, reject H_0. Otherwise, fail to reject H_0.
- A 95% (99%) "confidence level" in a test corresponds to $\alpha = 0.05$ (0.01).